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SUPERSEDING
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MILITARY SPECIFICATION

DOUGH MIXING MACHINE CONTINUOUS (WITH LIQUID SPONGE BLENDING AND FERMENTATION)

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a continuous dough mixing and liquid sponge fermentation machine. The unit has a maximum output capacity of 300 pounds of bread dough per hour and includes processing equipment necessary to blend, ferment, and mix the ingredients of the dough and to continuously develop the product in the required form for final proofing or baking. Sweet dough, developed at a maximum rate to 200 pounds per hour (lb/hr), may also be produced by the unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-B-2316 - Bakery Equipment, Including Unit, Assemblies, Repair Parts and Tools, Preparation for Delivery of.
MIL-E-55301 - Electromagnetic compatibility.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7320

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Service

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1399 - Interface Standard for Shipboard Systems.
MIL-STD-1399/300 - Interface Standard for Shipboard Systems Section 300, Electric Power, Alternating Current (Metric).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

BAKING INDUSTRY SANITATION STANDARD COMMITTEE (BISSC)

No. 20 - Liquid Ferment and Continuous Mix Processing Equipment.

(Application for copies should be addressed to the Baking Industry Sanitation Standards Committee, 521 Fifth Avenue, New York. NY 10017.)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)

No. 45 - Electric Installations on Shipboard.

(Application for copies should be addressed to the Institute of Electrical and Electronic Engineers, Inc., 345 East 47th Street, New York, NY 10017.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS - Industrial Controls and Systems.
MG-1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

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Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The dough mixing and fermentation unit shall be designed to provide a continuous through-put of bread dough or sweet dough at the required consistency and form for final proofing and baking, and shall consist of the components specified herein arranged as a production assembly. Unit shall include, but not be limited to, equipment for blending and mixing of the liquid sponge, ingredient metering and incorporating devices, premix development apparatus, and distribution conveyor. The unit furnished shall be completed to the extent that, when installed and connected to the required power and water source, the unit will process the required doughs in accordance with specified requirements.

3.2 First article. When specified (see 6.2), the contractor shall furnish a dough mixing machine for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified. Stainless steel, where required, shall conform to ASTM A167, types 302, 304 and 316. Materials subject to contact with the product shall be of composition and finish equal to or better than required by BISSC No. 20.

3.4 Standard commercial product. The dough mixing machine shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the dough mixing machine being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

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3.6 Design. The machine shall be of the design whereby a liquid sponge is developed in chilled water jacketed tanks, the sponge is then pumped via transfer tubing to the incorporator (where ingredients, as required, are metered into the sponge and the remainder of the flour is added), then to the developer (where final development occurs), the extruder (where the dough is shaped), and finally to the cut-off mechanism where the dough is scaled and cut to the proper weight and shape for panning. When operated as prescribed by the manufacturer, the machine shall continuously mix, at specified capacities, sweet type and white bread type doughs, according to recipe, for the production of sweet type dough products and white bread type dough products of scaled weights varying from 1 ounce to 28 ounces. The end product resulting from dough produced by a machine of this design, after proper proofing and baking, shall be at least equivalent in all respects to the same type of end product made from dough produced by the conventional method (horizontal type dough mixer). The entire assembly shall be designed to comply with the requirements of BISSC No. 20. Screw threads and pipe threads shall be in the inch system in accordance with FED-STD-H28. The dough mixing and fermentation equipment shall be designed to prevent conditions hazardous to personnel or deleterious to equipment. The unit shall be readily accessible for service and cleaning. Lubricants, except edible lubricants, shall not be permitted to enter the product zone.

3.7 Performance. After the equipment has been arranged and set up for the production run and the liquid sponge has been blended and fermented in accordance with the manufacturer's recommendations, the unit shall be capable of producing bread dough for final proofing or baking at a continuous rate of up to 300 lb/hr or shall produce sweet dough at a continuous rate of up to 200 lb/hr. The finished product shall be deposited on the conveyor in the size, shape, and consistency required for the specific baking applications. Ingredients contained in the final mix shall be proportionally metered in the required quantity to produce a product of high end-quality, uniform in texture, and free of rope.

3.8 Construction. The dough mixing and fermentation equipment shall withstand strains, jars, and vibrations incidental to shipping, storage, installation, and service without being subject to defects or permanent deformation. Electrically exposed energized parts shall be insulated; moving parts shall be protected or guarded. All power-operated devices shall be controlled at a centralized control panel. Pipe, hose, tubing, and fittings shall be sized to provide the output capacity specified in 3.7, and all connections shall be leak-tight. Each component shall be readily positioned. Unless otherwise specified (see 6.2), all assemblies shall be supported by plain deck feet. The machine shall be constructed in three major components:

a. Liquid sponge fermentation tanks with transfer pumps.
b. Main ingredient mixing table assembly.
c. Developer - extruder - conveyor assembly.

3.8.1 Combination blending-and-fermentation tanks. The blending-and-fermentation tank shall be constructed of stainless steel (ASTM A167, type 304), shall include a cooling water jacket sized to maintain the liquid sponge temperature at approximately 75° Fahrenheit (F), and shall be fitted with a low-speed gate-type agitator, and a high-speed propeller-type mixer. The

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agitator and the mixer shall each be driven by a properly sized electric motor controlled from the main panel. The agitator and mixer shall be constructed of stainless steel, secured to a base which shall be mounted on the tank, and shall have a nominal speed of 23 revolutions per minute (rpm) and 1,200 rpm respectively. The tank shall be watertight, equipped with a two-section cover properly vented, and shall include necessary valves and piping for connection to the coolant and product piping. The complete tank shall be supported on adjustable legs constructed of stainless steel. Tank capacity shall be at least 30 gallons (liquid). Unless otherwise specified (see 6.2), three tanks shall be furnished. Provisions for positioning flexible hose in the tank top shall be supplied. Transfer of the liquid sponge from either tank shall be accomplished by the transfer pump. Each tank shall be fitted with a manual shutoff valve.

3.8.2 Liquid sponge transfer pump. The liquid sponge transfer pump shall be constructed of stainless steel and shall be of the positive displacement type. Pump shall be driven by a constant speed electric motor of proper speed and horsepower (hp) rating to transfer the sponge at required rate. Pump shall be easily dismantled for cleaning. Pump and motor shall be mounted on a stainless steel platform with legs of suitable height. Provision shall be made for bolting to the deck as required (see 3.8). Necessary piping, valves, and connections shall be supplied for connection to the blending and fermentation tank and the liquid sponge ingredient tank.

3.8.3 Table. The table, fitted with a stainless steel top and six adjustable legs, shall be designed to support, with no resulting deformation, the ingredient tank assembly, control panel, flour feeder, incorporator, heat exchanger, premix pump, and integral piping and electrical conduit for these assemblies. The table shall include provisions for mounting the water-chilled refrigeration unit specified in 3.10.1.

3.8.4 Ingredient tanks and pumps. Unless otherwise specified (see 6.2), the minimum liquid capacity for the nutrient tank shall be 3 gallons and minimum liquid capacity for the shortening tank shall be 1-1/2 gallons. Tanks shall be constructed of stainless steel (ASTM A167, type 304). Each tank shall be fitted with the necessary piping and shall be supplied with an ingredient metering pump constructed of stainless steel. The tank shall be designed to hold the shortening, shall be electrically heated and thermostatically controlled within a temperature range of 50°F. Each pump shall be driven by an electric motor of adequate hp and controlled by the control panel to operate within a speed range of 0 to 175 rpm. Drainage provisions shall be furnished for each tank. Sufficient piping shall be supplied to connect the liquid sponge pump to the heat exchanger and to connect the other pump discharge connections to the incorporator.

3.8.5 Heat exchanger. The heat exchanger constructed of stainless steel (ASTM A167, type 304), shall be designed to provide necessary cooling of the liquid sponge so that the maximum temperature of the liquid sponge entering the incorporator does not exceed 75°F. The unit shall be readily connected to the liquid sponge and coolant piping and easily dismantled for cleaning. A temperature gage, with a 3-inch minimum face diameter, graduated in degrees from 0° to 180°F, shall be mounted on the heat exchanger discharge piping.

3.8.6 Flour feeder. A volumetric-type, flour feeder, constructed of stainless steel (ASTM A167, type 304), complete with variable speed drive,

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shall be furnished to supply the remaining flour needed to complete the sponge dough mix. The feeder shall be powered by an electric motor of adequate hp and controlled at the control panel. The feeder shall deposit a predetermined quantity of flour into the incorporator tube without bridging or becoming jammed. The hopper portion of the feeder shall be sized to hold the amount of flour required to satisfy the performance requirements specified in 3.7.

3.8.7 Incorporator. The incorporator shall be made of stainless steel (ASTM A167, type 304) and shall be driven by a constant speed gear type electric motor at the required rpm to properly incorporate flour, sponge, and ingredients. The incorporator shall be fitted with a flour tube and shall include the required connections for installation of flexible hose to the ingredient supply and discharge piping to the premix pump. A pressure gage, with a 2-inch minimum diameter face, graduated from 0 to 60 pound per square inch gage, shall be installed on the discharge side on the incorporator. The pressure gage assembly shall be of the diaphragm-oil type. The incorporator shall be remotely controlled. The incorporator shall be so constructed as to provide rapid disassembly for cleaning purposes.

3.8.8 Premix pump. The sponge-dough mix shall be delivered to the developer by means of a positive displacement variable speed, premix pump. The pump shall be constructed of stainless steel (ASTM A167, type 316) and shall have an output speed range of 20 to 60 rpm. Sufficient piping shall be furnished for connection to the incorporator and the developer. The pump shall be electric-motor-driven.

3.8.9 Dough developer and extruder. The dough developer shall be base-mounted to the conveyor support structure and shall be sized to provide the dough output specified in 3.7. The unit shall divide the dough in a preset volume and extrude the final product in the consistency, shape, and size required. The dough developer shall be driven by a variable speed electric drive motor. Body and all parts of the developer shall be constructed of corrosion-resistant materials. Speed of developer shall be variable and adjustable in the range of 100 rpm to 300 rpm. A tachometer shall be provided to indicate the speed at which the developer is operating. A meter shall be provided to indicate the power in amperes, that is utilized during the varying rates of production and consistencies of doughs being developed. The developer shall be furnished with both a round orifice type extruder and a ribbon type extruder, shaped to produce a flat strip of dough approximately 1/4 inch in thickness and approximately 8 inches wide. The extruded dough shall be cut to size by means of an air-cylinder-operated cutter which shall be timed to vary the size of cut. Air for cutter operation shall be supplied by a compressor furnished with each machine (see 3.8.11). The final product shall be deposited on the conveyor at a rate consistent with the specified output capacity. The unit shall be remotely controlled at the control panel and shall be equipped with a local manual override which shall become disengaged when the unit is power-operated.

3.8.10 Conveyor. The conveyor body shall be constructed of stainless steel (ASTM A167, type 304) and dough product belt shall be made of corrosion-resistant material. Overall length of conveyor, including that below the developer, shall be approximately 12 feet 6 inches long and shall extend approximately 8 feet beyond the opening in the extruder. The dough product belt shall be powered by a variable speed electric motor of adequate hp to

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provide a speed range of 0.65 to 11.50 feet per minute. Belt width shall be sized to meet the output requirement of 3.7. Means shall be provided for taking up slack in the belt. The conveyor shall be independently supported on adjustable legs, and shall be equipped with retractable casters, and shall be so designed that the conveyor may be extended.

3.8.11 Compressor. The air compressor shall be of sufficient capacity at required pressure to operate the dough cutoff mechanism at maximum load. The compressor shall be driven by a properly sized electric motor. The compressor shall be designed or equipped to produce oil-free air. Antifriction bearings in both the compressor and motor shall be of the prelubricated, sealed-for-life type. The compressor shall be equipped with a safety valve, moisture drain, integral pulsation chamber, and aftercooler. If the air system requires that the compressor start against pressure, the compressor shall be equipped with a pressure and moisture unloader valve and a check valve. Suitable hose and fittings shall be provided for connecting the compressor to the cutoff mechanism.

3.9 Electrical.

3.9.1 General electrical requirements. All electrical wiring and components shall conform to NFPA No. 70. For land based installation, the machine shall be designed for operation on 208 volt (V), 60 Hertz (Hz), 3 phase electrical power, unless otherwise specified (see 6.2). When shipboard installation is specified (see 6.2), the machine shall be designed for operation on 440V, 60 Hz, 3 phase electrical power and in accordance with MIL-STD-1399, MIL-STD-1399/300, and IEEE No. 45. For shipboard installations, all electrical equipment shall be designed for operation on an ungrounded electrical system, but shall operate satisfactorily and without hazard to personnel or equipment with an accidental ground of any conductor. All power operated components shall be controlled at a centralized, splash-proof control panel. The panel shall be secured to the table. The panel shall be constructed of 14 gage minimum stainless steel (ASTM A167, type 304) and shall be so arranged that the panel top is not higher than 76 inches from the floor. Controls and instruments shall include the necessary ON-OFF switches for each component, a main disconnect, timer, metering controls, and indicating lights.

3.9.2 Electrical equipment. All electric motors shall conform to NEMA MG-1. Motor controllers shall conform to NEMA 1C-S. Electric motors shall have one of the following enclosure types, dripproof, totally enclosed, or totally enclosed fan cooled. Electric motors shall be protected against the entrance of dust and water by externally shielding the motor enclosure, or by using a motor having a suitable frame enclosure type. The power output of all electric motors shall be sufficient to provide continuous full load operation of the machine without exceeding individual motor temperature or hp ratings. Motor controllers shall be housed in NEMA 1CS-6 type 12, or better, enclosures. Alternating current motors shall be equipped with thermal overload protection devices of the manual reset type. Direct current motors shall be protected by appropriate fuses. The controls and switches shall be so arranged that all line leads shall be open in the OFF position and all lines to controls, switches, and instruments shall be open when the main switch is in the OFF position. Electrical equipment shall be designed for operation in an ambient temperature of 40° Celsius (C), except that the shipboard installation (see 6.2), electrical equipment shall be designed for operation in an ambient temperature of 50°C.

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3.10 Accessories. When specified (see 6.2), the following accessories shall be furnished.

3.10.1 Water chiller. A self-contained water chiller refrigeration unit designed to provide chilled water to the heat exchanger and combination blending and fermentation tank coolant jackets shall be furnished. The unit shall be designed to supply the necessary coolant to maintain the liquid sponge temperature at approximately 75°F. The unit shall be equipped with the necessary piping, valves, and fittings for direct connection to the components serviced. Inlet water connections shall be as specified (see 6.2). The unit shall be mounted on the substructure of the table specified in 3.8.3.

3.10.2 Conveyor extensions. A 4-foot conveyor extension shall be supplied for attachment to the conveyor specified in 3.8.10. The extension shall be compatible with the design of the existing conveyor. Provisions for mounting the extension shall be furnished. Power requirements shall be in accordance with the manufacturer's recommendations. The number of conveyor extensions to be furnished shall be as specified (see 6.2). When an extension(s) is specified, the belt length shall meet the requirements of the extended conveyor.

3.11 Identification marking. Identification shall be permanently and legibly marked directly on the dough mixing machine or on a corrosion-resisting metal plate securely attached to the dough mixing machine at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name, and trademark to be readily identifiable to the manufacturer.

3.12 Instruction plates. The dough mixing machine shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.13 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.14 Safety devices. Safety devices shall be in accordance with 3.8, and shall not impair the operating function of the protected component.

3.15 Lubrication and gearing. All moving parts shall be lubricated in accordance with the manufacturer's standard practice. Lubrication shall be precluded from entering the product zone in a positive manner.

3.16 Service parts and maintenance tools. Service parts, including seals, packing, hose, piping, and maintenance tools shall be furnished when specified in the contract (see 6.2).

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3.17 Radio-interference suppression. When specified (see 6.2), the electrical components specified herein shall be designed for radio-interference suppression in accordance with the requirements specified in MIL-E-55301.

3.18 Handbook of instructions and drawings. When specified see (6.2), the contractor shall furnish one copy of the manufacturer's standard handbook. Two sets of drawings which shall include electrical and piping diagrams and assembly of components shall also be furnished.

3.19 Finish. The complete assembly shall be treated and painted in accordance with the manufacturer's commercial finish. Exterior stainless steel shall not be painted, but shall have at least a No. 4 standard finish. The paint shall be free from runs, sags, orange peel or other defects.

3.20 Workmanship.

3.20.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.20.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.20.3 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.20.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.21 Servicing and restoration. Each unit tested shall be serviced and restored to a service condition equal to the original condition of the unit, neglecting nominal wear incurred during the tests. The restoration shall include lubrication, paint touchup or repainting, as required for delivery.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one dough mixing machine when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4 and the tests of 4.5.2. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5.1, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All dough mixing machines offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level S-4 and an Acceptable Quality Level (AQL) of 2.5 expressed in defects per hundred units.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-2 and an AQL of 2.5 defects per hundred units.

4.4 Examination. Each machine selected shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement shall constitute a defect.

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4.5 Tests.

4.5.1 Operating test. The dough mixing machine shall be assembled and set up in a space available to the contractor. After assembly and connection to electrical and water supplies, each power-operated component shall be tested, and the machine shall be operated for 1 hour, without load, to determine that all parts function properly and that motors and bearings do not overheat. Each functional failure shall constitute a defect.

4.5.2 Performance test. The complete dough mixing and fermentation unit shall be operated for a production run of bread dough to determine compliance with 3.7. Ingredients shall be as recommended by the manufacturer except that the amount of flour used in the liquid sponge shall be 50 percent of the total flour used in the bread dough. After the initial blending and fermentation of the liquid sponge, the unit shall be operated continuously for an 8-hour period. Temperature of the liquid sponge, pressure of the sponge dough, amount of ingredients used, and mixing and fermentation time shall be recorded. The completed sponge dough shall be cut into preset round cross section quantities. The finished dough shall be inspected for compliance with 3.7. Upon completion of the test, the assembly shall be dismantled and checked for defects and ease of cleaning. Inability to maintain a continuous production rate of quality dough as specified in 3.7 and malfunction of any component shall constitute a defect for each failure.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-B-2316.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-B-2316 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. The unit specified herein is intended for bakeries and galleys to produce dough. Operation of the unit involves use of a liquid sponge mix which has been partially fermented. The unit, by means of a continuous process, is capable of producing dough ready for baking or proofing.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a dough mixing machine for first article inspection and approval is required and shall be furnished (see 3.2, 4.2.1, and 6.4).
c. When bolt-down deck feet are required (see 3.8).
d. Number of combination blending and fermentation tanks required, if different (see 3.8.1).
e. Number of ingredient tanks required, if different (see 3.8.4).
f. Electrical power requirements, if different (see 3.9.1).
g. When shipboard installation is required (see 3.9.1 and 3.9.2).
h. Accessories required (see 3.10).
i. Size of water inlet connection to the water chiller when required (see 3.10.1).

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j. Number of 4-foot conveyor extensions, when required (see 3.10.2).
k. When fungus treatment is required (see 3.13).
l. When service parts and maintenance tools is required (see 3.16).
m. When radio-interference suppression is required (see 3.17).
n. When handbook of instructions and drawing is required (see 3.18).
o. Level of preservation and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Options. When coolant provisions are not specified to be furnished with the combination blending and fermentation tanks and heat exchanger, the procuring activity must supply facilities for coolant in order to maintain the liquid sponge temperature within acceptable limits. Type and quantity of coolant used should be in accordance with manufacturer's recommendations.

6.6 System layout. In view of the extensiveness of the components which comprise the unit specified herein, layout of the various subassemblies should be made in accordance with the manufacturer's recommendations. When limitations of available space do not permit arrangement of the components in accordance with the manufacturer's intended layout, additional piping and wiring may be required.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

7320-N736

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-82085B(YD)	DOUGH MIXING MACHINE CONTINUOUS (WITH LIQUID SPONGE BLENDING AND FERMENTATION

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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